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Starkville, MS

# Kofi A. Ennin

## Education

**Mississippi State University, Shackouls Honors College** Expected May 2028

B.S. Computer Science, Artificial Intelligence concentration. GPA 3.71/4.0, major GPA 3.87/4.0. President's and Dean's Scholar.

*Coursework: Systems Programming, Computer Organization, Microprocessors, Data Structures & Algorithms (Honors), Digital Devices, Software Development Methods & Tools, Off-Road Autonomy, Discrete Structures, Linear Algebra, Calculus III; in progress: AI Robotics (SLAM, Path Planning, ROS), Probability.*

## Publications

- **K. A. Ennin** et al. OREAM: Off-Road End-to-End Autonomous Model for Unstructured Environments. *In preparation for SPIE.*
- Fortier et al. (incl. **K. A. Ennin**). Benchmarking Affordance Generalization with BusyBox. *CoRL.*
- Dabbiru et al. (incl. **K. A. Ennin**). CaSSeD: CAVS Semantic Segmentation Dataset for Off-Road Autonomous Driving.

## Research & Robotics

**MSU Center for Advanced Vehicular Systems (CAVS), Undergraduate Researcher** Oct 2024 – Present

- Leading development of OREAM, an end-to-end off-road planning model (PyTorch, CUDA): a TransFuser-based camera + LiDAR fusion backbone generates multiple anchor-based candidate trajectories, then selects one by combining a learned drivability cost, a LiDAR traversability cost, and a destination cost; built data and evaluation pipelines over camera and LiDAR driving logs.
- Benchmarking multi-trajectory planning heads, including winner-takes-all multi-mode decoders (MTP-style) and vocabulary-scoring heads (VADv2 / Hydra-MDP-style); integrating the planner into the MAVS autonomy stack and evaluating cross-season generalization, runtime, and accuracy against existing traversability models.
- Built a C++17 LiDAR pipeline that estimates tree-trunk diameter from off-road point clouds collected on a Warthog UGV: elevation-grid ground removal, a hand-written k-d tree for Euclidean clustering, and RANSAC and least-squares circle fitting (PCL, Eigen), with per-stage latency profiling.
- Contributed to NATURE, CAVS's open-source C++ ROS/ROS 2 autonomy stack for off-road navigation, extending its elevation-grid perception module and point-cloud interfaces.
- Adapting NVIDIA AlpaSim, an open-source closed-loop AV simulator, for evaluating off-road driving policies.
- Trained a FrustumNet model for 7-class semantic segmentation of off-road LiDAR point clouds on an expanded 450-scan labeled dataset (70/20/10 split), reaching 0.65 mIoU on held-out scans and 0.94 IoU on high vegetation.

**AI for Industry Challenge, Intrinsic × Open Robotics** 2026

- Built and validated a learned manipulation policy across ROS, Gazebo, MuJoCo, Isaac Sim, and real UR5e hardware, iterating through simulation and hardware testing toward deployment on a physical workcell.

**Microsoft Research, Robotics Research Intern** Aug 2025

- Automated data-collection pipelines and operated a robotic arm for BusyBox, a physical benchmark for affordance generalization in vision-language-action (VLA) models, with a 1,993-demonstration bimanual Mobile ALOHA dataset and $\pi_{0.5}$ and GR00T-N1.6 baselines; co-authored the resulting paper, accepted at CoRL.

**Autonomous Navigation, Mississippi State University** 2025

- Implemented and debugged SLAM, path planning, and state estimation in ROS with real-time sensor streams on TurtleBot 3 and Unitree Go1; root-caused failures across the hardware/software stack.

## Industry Experience

**Microsoft, Software Engineering Intern, Microsoft Sentinel** May 2026 – Aug 2026

*Redmond, WA*

- Designed and built Synthetics for Data Connectors, an automated diagnostic pipeline that validates production data ingestion end to end and flags schema drift, missing fields, and upstream API changes (TypeScript/Node.js, Docker, Azure CLI, MCP).
- Built connector-agnostic integration and regression test infrastructure with automated failure triage and reporting, architected to scale validation across hundreds of data connectors.
- Root-caused a Salesforce schema regression that existing monitoring missed; validated the framework against Salesforce, CrowdStrike, Trellix, and Atlassian connectors.

**Microsoft, Software Engineering Intern (Explore), Cloud Ecosystem Security** May 2025 – Aug 2025

*Redmond, WA*

- Worked with a three-person engineering team to scope and build agentic MCP developer tooling that converts hardcoded Azure Function tasks into declarative JSON ARM templates, cutting a two-week manual workflow to one day.

## Projects

**Vision-Based Robot Teleoperation, HackGT 13, Georgia Tech** 2026

- Built a markerless teleoperation prototype with a three-person team, aimed at remote robot-assisted surgery: browser-based MediaPipe pose tracking from a webcam drives an SO-101 arm in real time through a Python control bridge, with no leader arm needed.
- Engineered safety behaviors for tracking loss: one-second pose-hold to engage, hold on tracking loss, automatic resume, and return-to-home after 10 s with no operator in view.

**Campus Building Classifier, MSU AI Club Campus Vision Challenge (1st place)** 2024

- Fine-tuned a ResNet-50 in PyTorch to classify MSU campus buildings from photos, reaching 99.45% validation accuracy (F1 0.9945); built a GPU/CPU inference script for judging.

**SALO, AI Reading Assistant** 2025

- Desktop PDF reader (Electron, React, pdf.js, Tailwind) with a FastAPI backend streaming page summaries and chat over SSE; hybrid retrieval (Pinecone vector search + BM25) and Claude tool calls, with Firebase auth.

**Danso, Adaptive Study Platform for Ghana's NSMQ** 2024

- RAG study platform: FAISS retrieval over sentence-transformer embeddings feeding a quantized open LLM served with FastAPI, a Streamlit quiz UI, and a LinUCB contextual bandit that adapts questions to each learner.

## Technical Skills

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|---|---|
| **Languages** | C++, Python, C, TypeScript, JavaScript, C#, Assembly |
| **Robotics** | ROS 1/2, LeRobot, SO-101, NVIDIA AlpaSim, PCL, Eigen, Gazebo, MuJoCo, Isaac Sim, Sim-to-Real, LiDAR, Cameras, UR5e, Arduino |
| **Data & ML** | PyTorch, CUDA, NumPy, Data Ingestion & Validation, Schema Monitoring, Data Pipelines |
| **Testing** | Integration, Regression & Unit Testing, Test Automation, CI/CD, Failure Triage, Code Review |
| **Infrastructure** | Linux, Docker, Git/GitHub, Azure CLI, PowerShell, ARM Templates, MCP, RPC-style Tooling |

## Leadership & Awards

- Taught and onboarded 100+ high-school students in robotics; gathered feedback to iterate on technical guides and hands-on workshops (VEX, EV3, Arduino). MSU AI Club Team Lead; MSU Boxing Club Board Member; NSBE member.
- Mentor, Africa Deep Tech Challenge 2025: mentored teams in a pan-African competition on resource-constrained computing (2,800+ registrants, 400+ projects, $20K in prizes), powered by Future Africa with CcHUB, IHS Towers, and Ilorin Innovation Hub.
- 1st Place: MSU AI Club Campus Vision Challenge (2024); Robofest, Lawrence Tech (2023); RiSe National Robotics, Ghana (2022).